Answer Key

Other (142 Questions)

Q1. (B)	Q2. (C)	Q3. (D)	Q4. (C)	Q5. (B)
Q6. (C)	Q7. (C)	Q8. (C)	Q9. (B)	Q10. (D)
Q11. (B)	Q12. (D)	Q13. (B)	Q14. (C)	Q15. (C)
Q16. (C)	Q17. (B)	Q18. (A)	Q19. (C)	Q20. (C)
Q21. (A)	Q22. (B)	Q23. (A)	Q24. (C)	Q25. (D)
Q26. (A)	Q27. (A)	Q28. (B)	Q29. (A)	Q30. (D)
Q31. (C)	Q32. (D)	Q33. (D)	Q34. (B)	Q35. (B)
Q36. (A)	Q37. (C)	Q38. (A)	Q39. (D)	Q40. (A)
Q41. (D)	Q42. (B)	Q43. (C)	Q44. (C)	Q45. (D)
Q46. (B)	Q47. (D)	Q48. (C)	Q49. (B)	Q50. (A)
Q51. (B)	Q52. (B)	Q53. (B)	Q54. (C)	Q55. (C)
Q56. (A)	Q57. (D)	Q58. (C)	Q59. (C)	Q60. (A)
Q61. (C)	Q62. (B)	Q63. (B)	Q64. (B)	Q65. (A)
Q66. (B)	Q67. (C)	Q68. (B)	Q69. (A)	Q70. (D)
Q71. (A)	Q72. (D)	Q73. (B)	Q74. (B)	Q75. (B)
Q76. (A)	Q77. (C)	Q78. (C)	Q79. (D)	Q80. (A)
Q81. (A)	Q82. (C)	Q83. (B)	Q84. (C)	Q85. (C)
Q86. (C)	Q87. (B)	Q88. (B)	Q89. (D)	Q90. (A)
Q91. (B)	Q92. (B)	Q93. (B)	Q94. (C)	Q95. (D)
Q96. (C)	Q97. (A)	Q98. (C)	Q99. (B)	Q100.(C)
Q101.(A)	Q102.(D)	Q103.(A)	Q104.(B)	Q105.(B)

Q106.(C)	**Q107.(B)**	**Q108.(C)**	**Q109.(B)**	**Q110.(A)**
Q111.(C)	**Q112.(B)**	**Q113.(A)**	**Q114.(D)**	**Q115.(C)**
Q116.(D)	**Q117.(D)**	**Q118.(C)**	**Q119.(A)**	**Q120.(B)**
Q121.(D)	**Q122.(A)**	**Q123.(B)**	**Q124.(D)**	**Q125.(B)**
Q126.(A)	**Q127.(D)**	**Q128.(A)**	**Q129.(A)**	**Q130.(D)**
Q131.(A)	**Q132.(A)**	**Q133.(B)**	**Q134.(A)**	**Q135.(C)**
Q136.(A)	**Q137.(C)**	**Q138.(C)**	**Q139.(C)**	**Q140.(C)**
Q141.(B)	**Q142.(A)**			

Solutions

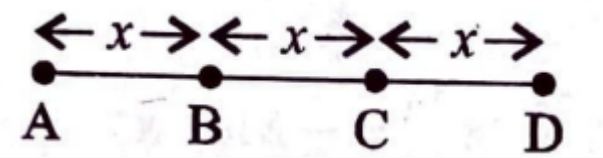
Q1. Solution

Correct Answer: (B)

Consider,

$\mathrm{AB} = x$

$\mathrm{BC} = x$



$2x + \mathrm{CD} = 3x \Rightarrow \mathrm{CD} = 3x - 2x = x$

Average speed of the object $< \mathrm{v} >$

$= \frac{\text{Total distance}}{\text{Total time}}$

$< \mathrm{v} > = \frac{3x}{\frac{x}{v_1} + \frac{x}{v_2} + \frac{x}{v_3}} = \frac{3v_1v_2v_3}{v_2v_3 + v_1v_3 + v_1v_2}$

Q2. Solution

Correct Answer: (C)

When temperature increases, more electrons excite to conduction band and hence conductivity increases, therefore resistance decreases.

Q3. Solution

Correct Answer: (D)

$\mathrm{N} = \mathrm{N}_0 \left(\frac{1}{2}\right)^{\mathrm{n}}$

$\mathrm{N} = \frac{\mathrm{N}_0}{8}$

$\frac{\mathrm{N}_0}{8} = \mathrm{N}_0 \left(\frac{1}{2}\right)^{\mathrm{n}} \Rightarrow \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{\mathrm{n}}$

$\mathrm{n} = 3$

3 half lives $= 3$ days

1 half life $= 1$ day

5 days $= 5$ half life

$\mathrm{N} = \mathrm{N}_0 \left(\frac{1}{2}\right)^{\mathrm{n}} \Rightarrow 8 \times 10^{-3} = \mathrm{N}_0 \left(\frac{1}{2}\right)^5$

$\Rightarrow \quad N_0 = 2^5 \times 8 \times 10^{-3} = 256\mathrm{gm}$

Q4. Solution

Correct Answer: (C)

If we assume electron in hydrogen atom takes energy $12.09\mathrm{eV}$ from the incoming radiation, the maximum excited state of electron will be $n = 3$. So, number of spectral lines is $\frac{3(3-1)}{2} = 3$.

Here we assume some part of energy $12.5\mathrm{eV} - 12.09\mathrm{eV} = 0.41\mathrm{eV}$ get lost due to collision.

Q5. Solution

Correct Answer: (B)

We have $V_f = V_i$

$\Rightarrow \frac{4}{3}\pi r_f^3 = 1000 \times \frac{4}{3}\pi r_i^3 \Rightarrow r_f^3 = 1000 r_i^3$

$\Rightarrow r_f = 10 r_i$

So, released energy = Initial surface energy - final surface energy

$= 1000 \times T \times 4\pi r_i^2 - T \times 4\pi r_f^2$

$= 4\pi T\left(1000 r_i^2 - r_f^2\right)$

$= 4\pi \times 0.07\left(1000 r_i^2 - 100 r_i^2\right)$

$= 4\pi \times 0.07 \times 900 r_i^2$

$= 4\pi \times 63 \times 10^{-6} = 7.92 \times 10^{-4}$ J

Q6. Solution

Correct Answer: (C)

Here, $q_1 = 1 \times 10^{-7}$C, q_2 and 2×10^{-7}C, $r = 20$ cm $= 20 \times 10^{-2}$ m

$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2} = \frac{9\times10^9\times1\times10^{-7}\times2\times10^{-7}}{\left(20\times10^{-2}\right)^2}$

$= 4.5 \times 10^{-3} N$

Q7. Solution

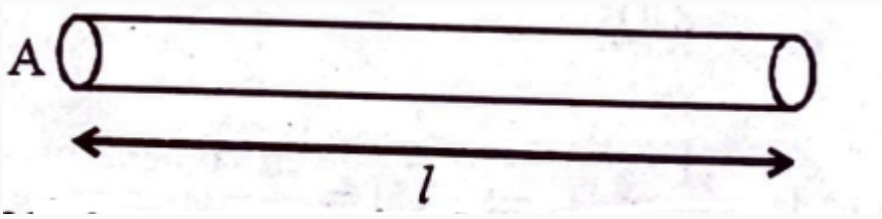
Correct Answer: (C)

Work done $= \frac{1}{2}\frac{q^2}{C} = \frac{\left(8\times10^{-18}\right)^2}{2\times100\times10^{-6}}$

$= 32 \times 10^{-32}$ J

Q8. Solution

Correct Answer: (C)



Let resistance of a wire R and length 1 .

$R = \frac{\rho\ell}{A} = 5\Omega$

$\therefore$ Volume of wire is constant in stretching

$V_i = V_f \Rightarrow A_i\ell_i = A_f\ell_f$

$A\ell = A'(5\ell) \Rightarrow A' = \frac{A}{5}$

$R_f = \frac{\rho\ell_f}{A_f} = \frac{\rho(5\ell)}{\left(\frac{A}{5}\right)} = 25\left(\frac{\rho\ell}{A}\right) = 25 \times 5 = 125\Omega$

Q9. Solution

Correct Answer: (B)

Given that uniform magnetic field, $\vec{B} = (2\hat{i} + 3\hat{j})T$

Acceleration $\vec{a} = (\alpha\hat{i} - 4\hat{j})\mathrm{m/s^2}$ We know that

$F = q(\vec{v} \times \vec{B}) \Rightarrow m\vec{a} = q(\vec{v} \times \vec{B})$

Here, $\vec{a} \perp \vec{B}$, so, $\vec{a} \cdot \vec{B} = 0$

$(\alpha\hat{i} - 4\hat{j})(2\hat{i} + 3\hat{j}) = 0 \Rightarrow 2\alpha - 12 = 0 \Rightarrow \alpha = 6$

Q10. Solution

Correct Answer: (D)

De Broglie wavelength is $\lambda = \frac{\mathrm{h}}{\mathrm{mv}}$

$\lambda_{\mathrm{p}} = \lambda_{\mathrm{e}} \Rightarrow \frac{\mathrm{h}}{\mathrm{m_p v_p}} = \frac{\mathrm{h}}{\mathrm{m_e v_e}}$

$\mathrm{m_e v_e} = \mathrm{m_p v_p} \Rightarrow \mathrm{p_e} = \mathrm{p_p}$

$\therefore \frac{\mathrm{p_p}}{\mathrm{p_e}} = \frac{1}{1}$

Q11. Solution

Correct Answer: (B)

Work done $=$ Area under the curve

$\Rightarrow \mathrm{W} = \frac{1}{2} \times (4 - 2) \times (400 - 100) = \frac{1}{2}(2) \times 300$

$\mathrm{W} = 300\ \mathrm{J}$ ~

Q12. Solution

Correct Answer: (D)

(d) To redirect the light that enters the telescope to the eyepiece or camera. The primary mirror of a reflecting telescopes gathers the light and reflects towards the secondary mirror which then reflect the light towards the eyepiece allowing the observer to see image.

It has advantage of a large focal length in a short telescope. ,

Q13. Solution

Correct Answer: (B)

As $\overrightarrow{\mathrm{M}} = \mathrm{IA}$

$\Rightarrow |\overrightarrow{\mathrm{M}}| = \frac{\mathrm{e}}{\frac{2\pi\mathrm{R}}{\mathrm{v}}}\pi\mathrm{R}^2 \left[\because \mathrm{I} = \frac{\mathrm{Q}}{\mathrm{T}} = \frac{\mathrm{e}}{\frac{2\pi\mathrm{R}}{\mathrm{V}}}\right]$

$\Rightarrow |\vec{M}| = \frac{1}{2}evR \Rightarrow |\vec{M}| = \frac{\mathrm{mvR}}{1} \cdot \frac{\mathrm{e}}{2\,\mathrm{m}}$

$\Rightarrow |\overrightarrow{\mathrm{M}}| = \frac{\mathrm{eL}}{2\,\mathrm{m}} \Rightarrow |\overrightarrow{\mathrm{M}}| = -\frac{\overrightarrow{\mathrm{eL}}}{2\,\mathrm{m}}$

[$\because$ Here $\overrightarrow{\mathrm{M}}$ and $\overrightarrow{\mathrm{L}}$ will always be opposite] ~

Q14. Solution

Correct Answer: (C)

Intensity at a point in Young's double slit experiment is given by

$\mathrm{I}=\mathrm{I}_1+\mathrm{I}_2+2\sqrt{\mathrm{I}_1\mathrm{I}_2}\cos\phi$

Here $\mathrm{I}_1=\mathrm{I}_2=\mathrm{I}_0$ (say)

At P

$\therefore \mathrm{I}_\mathrm{p}=\mathrm{I}_0+\mathrm{I}_0+2\mathrm{I}_0\cos\frac{\pi}{3}=2\mathrm{I}_0+2\mathrm{I}_0\times\frac{1}{2}=3\mathrm{I}_0$

At Q

$\mathrm{I}_\mathrm{Q}=\mathrm{I}_0+\mathrm{I}_0+2\mathrm{I}_0\cos 90^\circ=2\mathrm{I}_0$

$\frac{\mathrm{I}_\mathrm{P}}{\mathrm{I}_\mathrm{Q}}=\frac{3}{2}$,

Q15. Solution

Correct Answer: (C)

From the ideal gas equation $\mathrm{PV}=\mathrm{nRT}$ Here, volume is constant $\therefore \frac{\mathrm{P}_1}{\mathrm{T}_1}=\frac{\mathrm{P}_2}{\mathrm{T}_2}$

Here, $T_1=27+273=300\ \mathrm{K}$

$P_1=270\mathrm{kPa}$

$\mathrm{T}_2=36+273=309\ \mathrm{K}$

$\Rightarrow P_2=\frac{P_1}{T_1}\times T_2=\frac{270\times(309)}{300}=278\mathrm{kPa}$.

Q16. Solution

Correct Answer: (C)

Let the distance from the mean position is X.

Given $\mathrm{KE}=\mathrm{PE}$

So, $\frac{1}{2}M\omega^2\left(A^2-x^2\right)=\frac{1}{2}M\omega^2x^2$

$A^2-x^2=x^2\Rightarrow A^2=2\times 2$

$\therefore \mathrm{x}=\pm\frac{\mathrm{A}}{\sqrt{2}}$ ^

Q17. Solution

Correct Answer: (B)

Electric field in a certain region is given by,

$\vec{E}=\frac{A}{x^2}\hat{i}+\frac{B}{y^3}\hat{j}$

$\left[\frac{\mathrm{A}}{\mathrm{x}^2}\right]=\mathrm{NC}^{-1}\Rightarrow[\mathrm{A}]=\mathrm{Nm}^2\mathrm{C}^{-1}$

$\left[\frac{\mathrm{B}}{\mathrm{y}^3}\right]=\mathrm{NC}^{-1}\Rightarrow[\mathrm{B}]=\mathrm{Nm}^3\mathrm{C}^{-1}$ ^

Q18. Solution

Correct Answer: (A)

Mass of particle, $\mathrm{m} = 500\ \mathrm{g} = 0.5\ \mathrm{kg}$

velocity of a particle, $\vec{v} = 2t\hat{i} + 3t^2\hat{j}$

$\overrightarrow{\mathrm{a}} = \frac{\mathrm{d}\vec{\mathrm{v}}}{\mathrm{dt}} = 2\hat{\mathrm{i}} + 6\mathrm{t}\hat{\mathrm{j}}$

at $\mathrm{t} = 1, \overrightarrow{\mathrm{a}} = 2\hat{\mathrm{i}} + 6\hat{\mathrm{j}}$

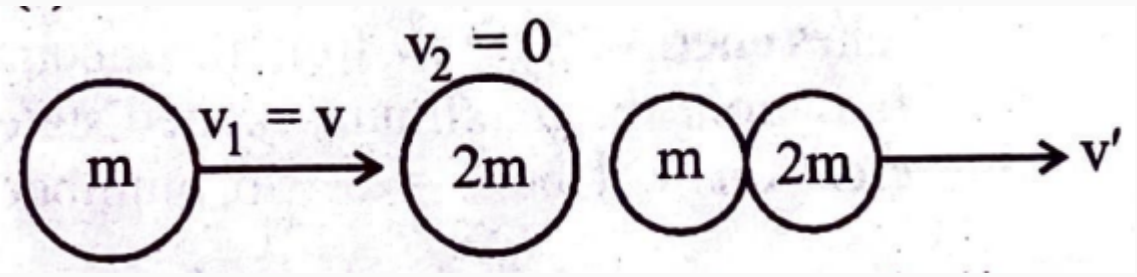
Force acting on the particle,

$\overrightarrow{\mathrm{F}} = \mathrm{ma} = 0.5(2\hat{\mathrm{i}} + 6\hat{\mathrm{j}}) = \hat{\mathrm{i}} + 3\hat{\mathrm{j}}$

$\overrightarrow{\mathrm{F}} = \hat{\mathrm{i}} + x\hat{\mathrm{j}}$ Hence $x = 3$ ^

Q19. Solution

Correct Answer: (C)



Applying conservation of linear momentum

$\Rightarrow \overrightarrow{\mathrm{P}}_{\mathrm{i}} = \overrightarrow{\mathrm{P}}_{\mathrm{f}} (\because \mathrm{P} = \mathrm{mv})$

$mv_1 + 2mv_2 = (m + 2m)v'$

$mv^1 + 2m \times 0 = (3m)v'$

$\Rightarrow \mathrm{mv} = 3\mathrm{mv}' \Rightarrow \mathrm{v}' = \frac{\mathrm{v}}{3}$!

Q20. Solution

Correct Answer: (C)

For time varying condition Maxwell's equation, $\oint \overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{dl}} = -\frac{\mathrm{d}\phi_{\mathrm{B}}}{\mathrm{dt}}$,

Q21. Solution

Correct Answer: (A)

The resonance frequency of LC oscillations circuit is

$\omega = \frac{1}{\sqrt{\mathrm{L'C'}}} \Rightarrow \mathrm{L'} \rightarrow 2\ \mathrm{L}$

$\mathrm{C'} \rightarrow 8\mathrm{C}$

$\omega = \frac{1}{\sqrt{2\ \mathrm{L} \times 8\mathrm{C}}} = \frac{1}{4\sqrt{\mathrm{LC}}} 9\omega_0 = \frac{1}{\sqrt{\mathrm{LC}}}$

$\omega = \frac{\omega_0}{4}$ So, $x = \frac{1}{4}$

Q22. Solution

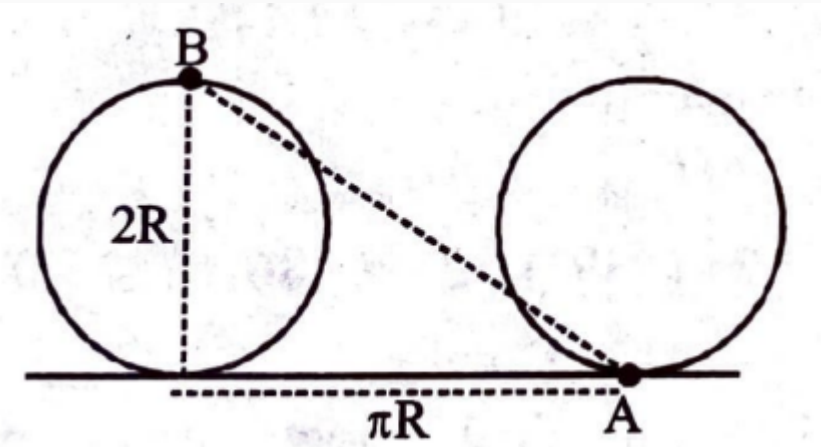
Correct Answer: (B)

As $\varepsilon|_{t=0.5\text{sec}} = -\frac{\mathrm{d}\phi}{\mathrm{dt}}$
$= -\mathrm{A}\frac{\mathrm{dB}}{\mathrm{dt}} \quad [\because \theta = 0^\circ \Rightarrow \cos\theta = 1]$
$= -\pi \times \left(\frac{10}{\sqrt{\pi}}\right)^2 \times 10^{-4} \times \frac{0-0.5}{0.5} = 10^{-2}\ \mathrm{V} = 10\mathrm{mV}$
As $\frac{dB}{dt}$ = constant $\Rightarrow$ Induced emf will not change with time. So, $\mathrm{e}|_{0,5\text{sec}} = \mathrm{e}|_{0.25\text{sec}} = 10\mathrm{mV}$

Q23. Solution

Correct Answer: (A)

(a) From figure,



Displacement,
$\mathrm{BA} = \sqrt{(2\mathrm{R})^2 + (\pi\mathrm{R})^2} = \mathrm{R}\sqrt{4+\pi^2}$

Q24. Solution

Correct Answer: (C)

Acceleration due to gravity,
$\mathrm{g} = \frac{\mathrm{GM}}{\mathrm{R}^2} = \frac{4}{3}\pi\mathrm{G}\rho\mathrm{R}$
$\therefore \frac{g_2}{g_1} = \frac{\rho_2}{\rho_1} \times \frac{R_2}{R_1} = \frac{1}{2} \times 1.5 = \frac{3}{4}$

Q25. Solution

Correct Answer: (D)

$\Delta\ell = \frac{F\ell}{YA} = \frac{250\times100}{10^{10}\times6.25\times10^{-4}} = 40\times10^{-4}\ \mathrm{m}$
$= 4\times10^{-3}\ \mathrm{m}$

Q26. Solution

Correct Answer: (A)

Given $\mathrm{M}_{\mathrm{H}_2}=2; \mathrm{M}_{\mathrm{O}_2}=32$

Speed of sound, $\mathrm{v}=\sqrt{\frac{\gamma \mathrm{RT}}{\mathrm{M}}}$

$\Rightarrow \quad v \propto \frac{1}{\sqrt{M}}$

$\therefore \quad \frac{\mathrm{v}_{\mathrm{H}_2}}{\mathrm{v}_{\mathrm{O}_2}}=\sqrt{\frac{\mathrm{M}_{\mathrm{O}_2}}{\mathrm{M}_{\mathrm{H}_2}}}=\sqrt{\frac{32}{2}}=4:1$

Q27. Solution

Correct Answer: (A)

Given,

Susceptibility of material, $\chi_m=2 \times 10^{-2}$

Using $\mu_{\mathrm{r}}=1+\chi_{\mathrm{m}}=1+0.02=1.02$

$\mathrm{B}_{\text{final}}=\mu_{\mathrm{r}} \mathrm{B}_0$ (here, $\mathrm{B}_0=$ initial magnetic field)

% increase in magnetic field

$=\frac{\mathrm{B}_{\text{final}}-\mathrm{B}_0}{\mathrm{B}_0} \times 100=\frac{\mu_{\mathrm{r}} \mathrm{B}_0-\mathrm{B}_0 \times 100}{\mathrm{B}_0}$

$=\frac{(\chi+1)-1 \times 100}{1}=0.02 \times 100=2\%$

Q28. Solution

Correct Answer: (B)

We have $\frac{\mathrm{U}_{\mathrm{E}}}{\mathrm{U}_{\mathrm{T}}}=\frac{\mathrm{U}_{\mathrm{E}}}{\mathrm{U}_{\mathrm{E}}+\mathrm{U}_{\mathrm{B}}}=\frac{\mathrm{U}_{\mathrm{E}}}{2 \mathrm{U}_{\mathrm{E}}}=\frac{1}{2}$

$\left[\because \mathrm{U}_{\mathrm{E}}=\mathrm{U}_{\mathrm{B}}=\frac{1}{2} \in_0 \mathrm{E}_0^2=\frac{\mathrm{B}_0^2}{2 \mu_0}\right]$

Q29. Solution

Correct Answer: (A)

Resultant intensity in Young's double slit experiment

$\mathrm{I}=4 \mathrm{I}_0 \cos^2\left(\frac{\Delta \phi}{2}\right)$

For path difference $\frac{\lambda}{4}$ phase difference,

$\Delta \phi=\frac{2 \pi}{\lambda} \times \frac{\lambda}{4}=\frac{\pi}{4}$

$\therefore \mathrm{I}_1=4 \mathrm{I}_0 \cos^2\left(\frac{\pi}{4}\right)=2 \mathrm{I}_0$

For path difference $\frac{\lambda}{3}$

$I_2=4 I_0 \cos^2\left(\frac{2 \pi}{\lambda} \times \frac{\lambda}{3}\right)=I_0$

$\therefore \frac{\mathrm{I}_1+\mathrm{I}_2}{\mathrm{I}_0}=3$

Q30. Solution

Correct Answer: (D)

As $\mathrm{E}(\mathrm{eV})=\frac{1240}{\lambda(\mathrm{nm})}=\frac{1240}{124.1} \simeq 10 \mathrm{eV}$

Only is transition (D), the energy gap is $10 \mathrm{eV}$ So, option (d) is correct

Q31. Solution
Correct Answer: (C)

Frenkel and Schottky defects are crystal defects. It arises due to dislodgement of cation or anion from their places in the crystal lattice.

Q32. Solution
Correct Answer: (D)

Given : Radius of hydrogen atom $= 0.530$
$\mathring{A}$, Number of excited state $(n) = 2$ and atomic number of hydrogen atom $(Z) = 1$. We know that the Bohr radius
$(r) = \frac{n^2}{Z} \times$ radius of atom $= \frac{(2)^2}{1} \times 0.530$
$= 4 \times 0.530 = 2.12\mathring{A}$

Q33. Solution
Correct Answer: (D)

The probability density of electrons in $2s$ orbital first increases then decreases and after that it increases again as distance increases from nucleus.

Q34. Solution
Correct Answer: (B)

Its valence shell has 5 electrons $\left(ns^2, np^3\right)$ It belongs to 15 th group of the periodic table.

Q35. Solution
Correct Answer: (B)

The stability of the ionic bond depends upon the lattice energy which is expected to be more between Mg and F due to +2 charge on Mg atom.

Q36. Solution
Correct Answer: (A)

7.5 mol
$\Delta T_f = K_f m$
$\Delta T_f = K_f \frac{n_2 \times 1000}{w_1}$
$\Rightarrow 14 = 1.86 \times \frac{n_2 \times 1000}{1000}$
$n_2 = 7.5$ mol

Q37. Solution

Correct Answer: (C)

$\Lambda_{\mathrm{m}} = 1000 \times \frac{\mathrm{K}}{\mathrm{M}}$
$= 1000 \times \frac{2 \times 10^{-5}}{0.001} = 20 \text{ S cm}^2 \text{ mol}^{-1}$
$\Rightarrow \alpha = \frac{\Lambda_{\mathrm{m}}}{\Lambda_{\mathrm{m}}^{\circ}} = \frac{20}{190} = \left(\frac{2}{19}\right)$

$$\underset{0.001(1-\alpha)}{\mathrm{HA}} \rightleftharpoons \underset{0.001\alpha}{\mathrm{H}^+} + \underset{0.001\alpha}{\mathrm{A}^-}$$

$K_{\mathrm{a}} = 0.001\left(\frac{\alpha^2}{1-\alpha}\right) = \frac{0.001 \times \left(\frac{2}{19}\right)^2}{1-\left(\frac{2}{19}\right)}$
$= 12.3 \times 10^{-6}$

Q38. Solution

Correct Answer: (A)

As per Arrhenius equation $\left(k = A\mathrm{e}^{-E_a/RT}\right)$, the rate constant increases exponentially with temperature.

Q39. Solution

Correct Answer: (D)

The correct answer is (4) All of the above; all three methods - mixing two oppositely charged sols, electrophoresis, and adding electrolytes - can be used for coagulation of a sol. Explanation: Mixing oppositely charged sols: When two sols with opposite charges are mixed, their charges neutralize each other, causing the particles to aggregate and precipitate out. Electrophoresis: Applying an electric field to a colloidal solution causes the charged particles to move towards the electrode with the opposite charge, where they lose their charge and coagulate. Adding electrolytes: Adding an electrolyte to a colloidal solution provides ions that can neutralize the charge on the colloidal particles, leading to coagulation.

Q40. Solution

Correct Answer: (A)

At $900 - 1500$ K (higher temperature range in the blast furnace)
Reaction which take place are:
$\mathrm{C} + \mathrm{CO_2} \longrightarrow 2\mathrm{CO}$
$\mathrm{FeO} + \mathrm{CO} \longrightarrow \mathrm{Fe} + \mathrm{CO_2}$
$\mathrm{CaO} + \mathrm{SiO_2} \longrightarrow \mathrm{CaSiO_3}$ (Slag formation)
$\mathrm{Fe_2O_3} + \mathrm{CO} \longrightarrow 2\mathrm{FeO} + \mathrm{CO_2}$ at $500 - 800$ K

Q41. Solution

Correct Answer: (D)

Kinetic theory of gases proves all the given gas laws.

Q42. Solution

Correct Answer: (B)

Enthalpy of formation of C_2H_4, CO_2 and H_2O are $52, -394$ and $-286\ \mathrm{kJ/mol}$ respectively.
(Given)
The reaction is
$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$
change in enthalpy,
$(\Delta H) = \Delta H_{\text{products}} - \Delta H_{\text{reactants}}$
$= 2 \times (-394) + 2 \times (-286) - (52 + 0)$
$= -1412\ \mathrm{kJ/mol}.$

Q43. Solution

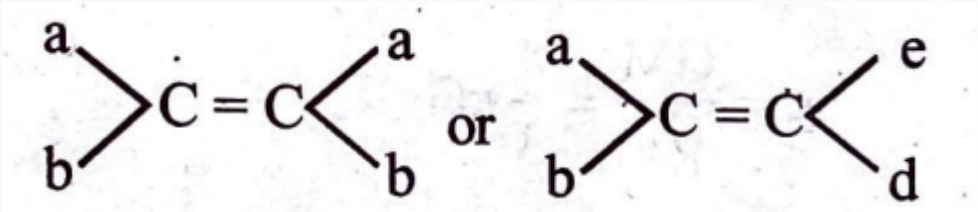
Correct Answer: (C)

Photochemical smog contains nitrogen dioxide (NO_2), Ozone (O_3), PAN (peroxyacetylnitrate), and compounds containing $-CHO$ group.

Q44. Solution

Correct Answer: (C)

The condition for geometrical isomerism is



$CH_2 = C(Cl)CH_3$ does not follow above mention condition.

Q45. Solution

Correct Answer: (D)

Separating funnel is used when the two liquids are immiscible.

Q46. Solution

Correct Answer: (B)

$2Al(s) + 2NaOH(aq) + 6H_2O(l) \rightarrow 2Na^+[Al(OH)_4]^-(aq) + 3H_2(\ g)$

Q47. Solution

Correct Answer: (D)

Down the group solubility of sulphate decreases. Thus, Ba^{2+} ions will precipitate out most easily.

Q48. Solution
Correct Answer: (C)

Ionic hydrides give the basic solution when it reacts with water, e.g.,
$\mathrm{LiH} + \mathrm{H_2O} \longrightarrow \mathrm{LiOH} + \mathrm{H_2}$

Q49. Solution
Correct Answer: (B)

Tofranil is used for the treatment of antidepressant.

Q50. Solution
Correct Answer: (A)

Melamine plastic crockery is a copolymer of HCHO and Melamine.

Q51. Solution
Correct Answer: (B)

The α-helix structure is formed when the chain of α-amino acids coils as a right handed screw (called α-helix) because of the formation of hydrogen bonds between amide groups of the same peptide chain, i.e., NH group in one unit is linked to carbonyl oxygen of the third unit by hydrogen bonding. This hydrogen bonding between different units is responsible for holding helix in a position.

Q52. Solution
Correct Answer: (B)

Inductive effect, steric hinderance and solvation effect the basic strength of amines.

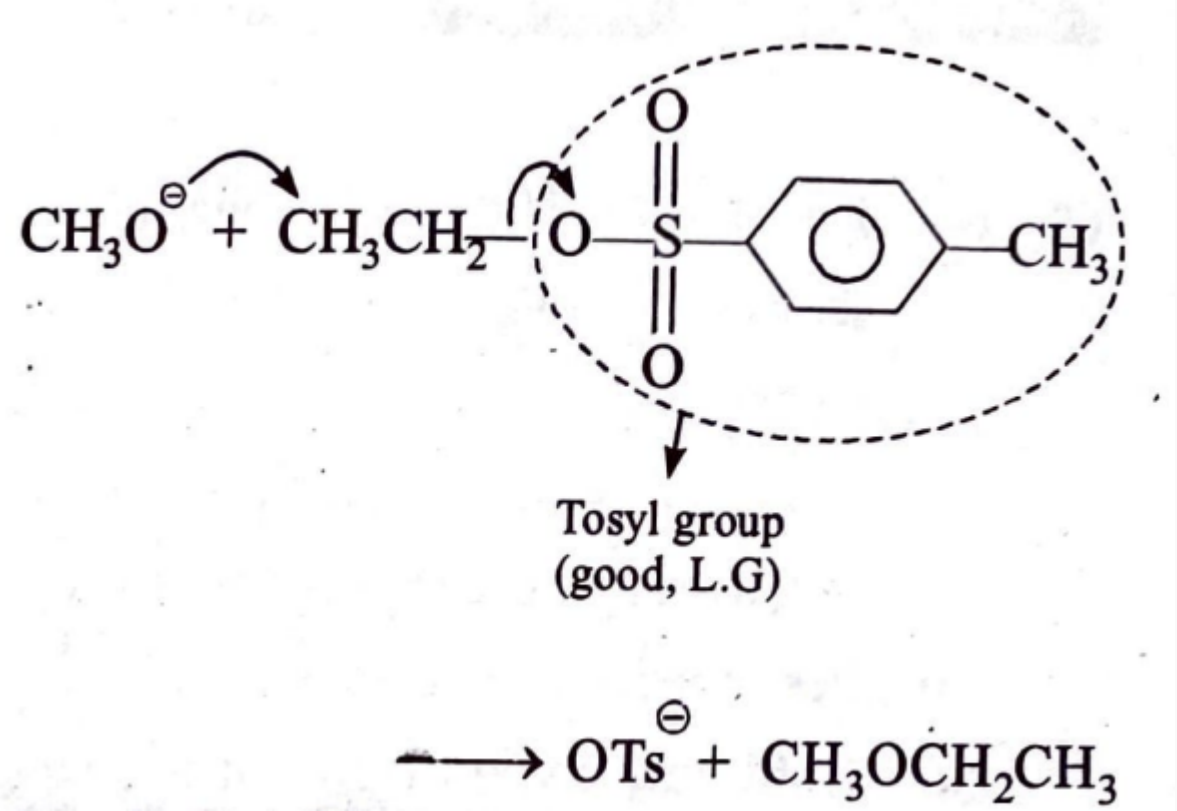
Q53. Solution
Correct Answer: (B)

Step 1: Friedel-Crafts Acylation Toluene undergoes acylation in the presence of acetyl chloride $(\mathrm{CH_3COCl})$ and $\mathrm{AlCl_3}$, leading to the formation of acetophenone. $\mathrm{C_6H_5CH_3} + \mathrm{CH_3COCl} \xrightarrow{\mathrm{AlCl_3}} \mathrm{C_6H_5COCH_3}$ Step 2: Partial Oxidation using $\mathrm{CrO_3}$ in Acetic Anhydride The acetophenone undergoes oxidation using $\mathrm{CrO_3}$ in $(\mathrm{CH_3CO})_2\mathrm{O}$, which converts the methyl group $(-\mathrm{CH_3})$ into an aldehyde (-CHO), forming benzaldehyde.
$\mathrm{C_6H_5COCH_3} \xrightarrow{\mathrm{CrO_3/(CH_3CO_2O,H_3O^+}} \mathrm{C_6H_5CHO}$ Overall Reaction:
$\mathrm{C_6H_5CH_3} \xrightarrow{\mathrm{AlCl_3,CH_3COCl}} \mathrm{C_6H_5COCH_3} \xrightarrow{\mathrm{CrO_3/(CH_3CO_2O,H_3O^+}} \mathrm{C_6H_5CHO}$ Thus, the final product B is benzaldehyde $(\mathrm{C_6H_5CHO})$.

Q54. Solution

Correct Answer: (C)

At 443 K compound in (d) will produce propene. In (a) alkene will be produced as tertiary halide and strong base favours elimination. In (b) reaction is not possibie at room temperature as due to resonance $C-Cl$ bond has partial double bond character which is very difficult to break.



Q55. Solution

Correct Answer: (C)

For the same product in both S_N1 and S_N2, the substitution must occur without rearrangement. - (I) 2-Bromo-2-methylpentane $\rightarrow$ S_N1 forms a tertiary carbocation, but S_N2 is hindered. (Different products possible, $\times$) - (II) 1-Chloro-1-methylcyclohexane $\rightarrow$ Possible carbocation rearrangement in S_N 1. (Different products possible, $\times$) - (III) Cyclohexyl chloride $\rightarrow$ Forms stable carbocation in S_N1 and undergoes backside attack in S_N2, leading to the same product. (✓) - (IV) 2-Bromobutane $\rightarrow$ Secondary halide, substitution leads to same product in both mechanisms. (✓) Correct Answer: (3) (III) and (IV).

Q56. Solution

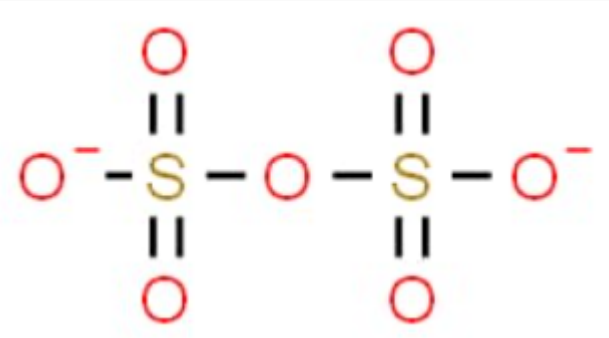
Correct Answer: (A)

Ligands can be arranged in a series in the orders of increasing field strength as given below:
Weak field ligands:
$I^- < Br^- < S^{2-} < SCN < Cl^- < N_3^-, F^- <$ Urea, OH $<$ oxalate
Strong field ligands
$O^{--} < H_2O < NCS^- < EDTA \ < Py, NH_3 <$ en $= SO_3^- <$bipy, Phen $< NO_2^- < CH_3^-$
$< C_6H_5^- < CN^- < CO$
Such a series is termed as spectrochemical series. It is an experimentally determined series based on the absorption of light by complexes with different ligands.

Q57. Solution

Correct Answer: (D)

$S_2O_7^{2-}$ has $S-O-S$ bond.



Q58. Solution

Correct Answer: (C)

In laboratory, manganese (II) ion salt is oxidised to permanganate ion in aqueous solution by peroxodisulphate.

$$2Mn^{2+} + \underset{\text{peroxodisulphate ion}}{5\ S_2O_8^{2-}} + 8H_2O \rightarrow 2MnO_4^- + 10SO_4^{2-} + 16H^+$$

Q59. Solution

Correct Answer: (C)

Boron in B_2H_6 is electron deficient

Q60. Solution

Correct Answer: (A)

	Fe	A1	B
E°_{Red}	0.77	-1.66	1.08
E°_{Oxi}	-0.77	1.66	-1.08

Hence, reducing power $Al > Fe^{2+} > Br$

Q61. Solution
Correct Answer: (C)

Fricassee refers to a method of cooking in which meat (often chicken or veal) is cut into pieces, lightly sautéed, and then stewed in a sauce, typically with vegetables.

Q62. Solution
Correct Answer: (B)

Retribution is punishment, contempt is feeling of disgust and grudge is an illfeeling.

Q63. Solution
Correct Answer: (B)

Sumptuous means sufficient or more in quantity, meagre means very little.

Q64. Solution
Correct Answer: (B)

Pertinent means relevant .

Q65. Solution
Correct Answer: (A)

Cold and couldn't go out are suggestive that it was very cold because of which we couldn't go out. So the word substitute for very is 'so' and because is 'that'.

Q66. Solution
Correct Answer: (B)

Faced with the challenge of the intrusion of colonial culture and ideology at attempt to reinvigorate traditional institutions and realize the potential of traditional culture in the pre-independence India developed during the nineteenth century.

Q67. Solution
Correct Answer: (C)

A diversified use of natural gas is emerging adding a new dimension to the traditional use of gas as a heating or power generation fuel by converting gas into amongst other products, high quality diesel transportation fuel virtually free of sulphur.

Q68. Solution
Correct Answer: (B)

Music is often linked to mood. A certain song can make us feel happy, sad, energetic, or relaxed.

Q69. Solution

Correct Answer: (A)

Music helps one feel different emotions. Based on the mood, a certain song can make us feel happy, sad, energetic, or relaxed.

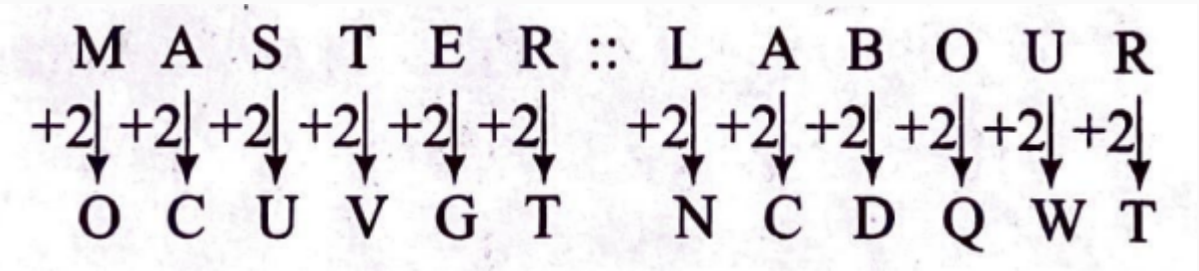
Q70. Solution

Correct Answer: (D)

All forms of music may have therapeutic effects, although music from one's own culture may be most effective.

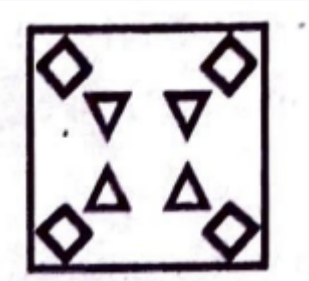
Q71. Solution

Correct Answer: (A)



Q72. Solution

Correct Answer: (D)



Q73. Solution

Correct Answer: (B)

In this code the alphabets are coded as follows

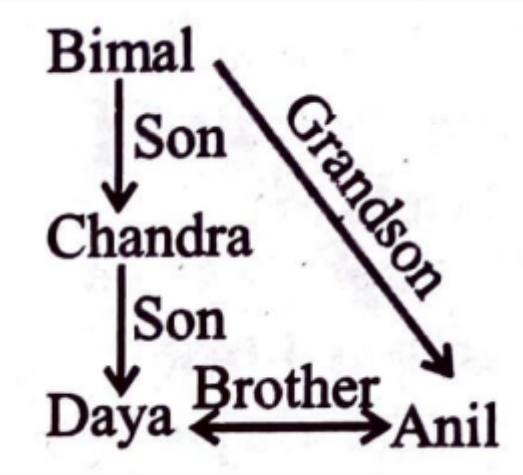
SISTER UNCLE BOY

535301 84670 129

If we apply this method, the code comes out to be 185336

Q74. Solution

Correct Answer: (B)



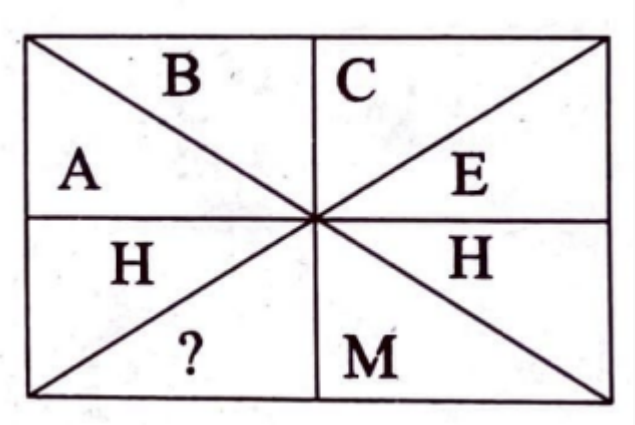
Q75. Solution

Correct Answer: (B)

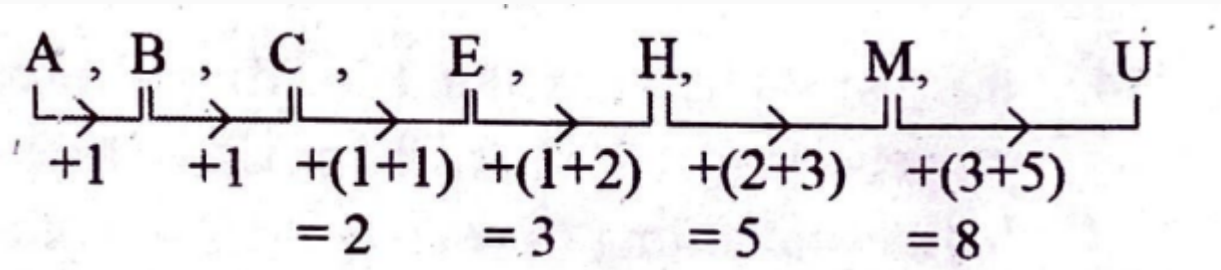
Except option (b) all given pairs are synonyms to each other.

Q76. Solution

Correct Answer: (A)



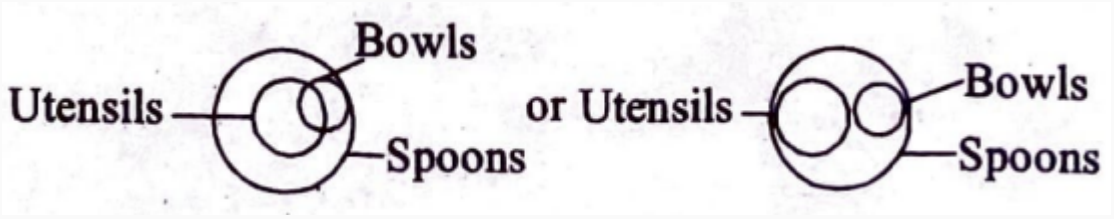
Pattern is, each next term is at the gap of sum of gaps of first two continious terms. Like:



Q77. Solution
Correct Answer: (C)

Flunchers, Flunching, Flunpites, Fluntlock, Fluntlocks

Q78. Solution
Correct Answer: (C)



Hence, either conclusion I or II follows.

Q79. Solution
Correct Answer: (D)

The word that cannot be formed using the letters of "CHEMOTHERAPY" is: (4) FATHER Explanation: Let's check each option: HECTARE → ✅ Can be formed (H, E, C, T, A, R, E are all present in "CHEMOTHERAPY") MOTHER → ✅ Can be formed (M, O, T, H, E, R are all present in "CHEMOTHERAPY") THEATER → ✅ Can be formed (T, H, E, A, T, E, R are all present in "CHEMOTHERAPY") FATHER → ❌ Cannot be formed (There is no F in "CHEMOTHERAPY") Thus, the correct answer is (4) FATHER.

Q80. Solution
Correct Answer: (A)

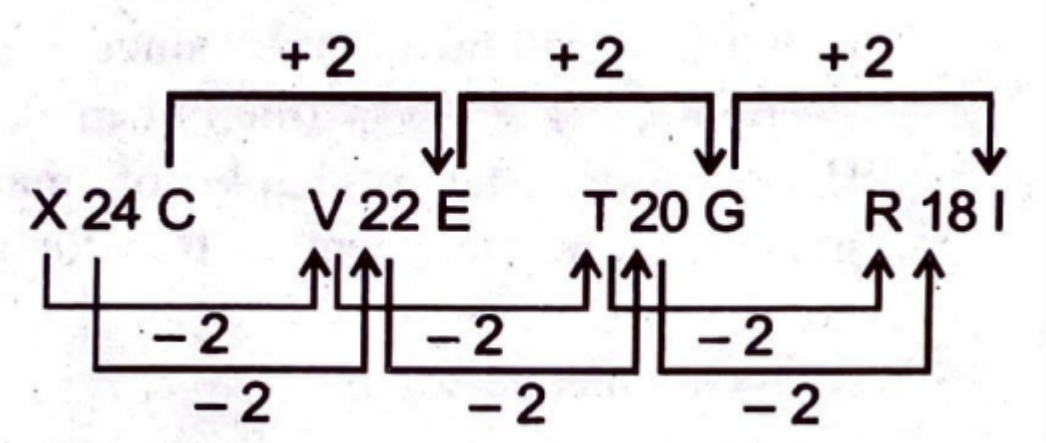
Inserting fggf in the blanks: fgg [fggf] gff [f] f [g] gfg [f] fgf0 Checking the continuity of the pattern: "fgg" appears twice. "gff" follows a similar alternating pattern. "fggf" maintains symmetry in the sequence. This seems to be the correct answer. Checking Other Options: (2) ccfc → Does not fit as "c" is not in the original sequence. ❌ (3) fgfg → Breaks the observed pattern. ❌ (4) ffgg → Disrupts the alternating pattern. ❌

Q81. Solution
Correct Answer: (A)

As, $(9, 217, 8)$
$9 \times 8 \Rightarrow 72 \times 3 + 1 = 217$
Similarly, $(4, 37, 3)$
$4 \times 3 = 12 \times 3 + 1 = 37$

Q82. Solution

Correct Answer: (C)



Q83. Solution

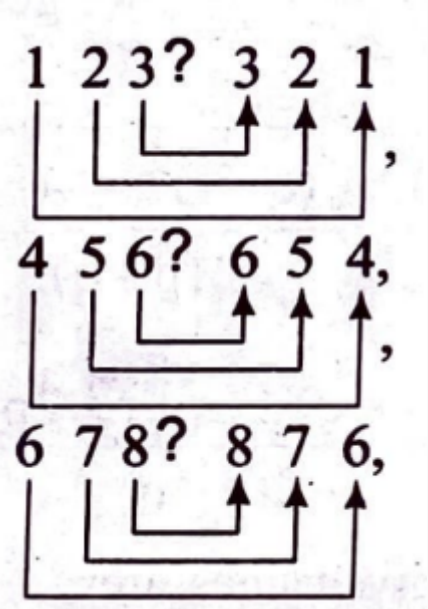
Correct Answer: (B)

$108^2 = 11664$

Similarly, $112^2 = 12544$

Q84. Solution

Correct Answer: (C)



but $789 - 978$ does not follow this rule.

Q85. Solution

Correct Answer: (C)

The pattern followed is: $\rightarrow 3+4=7\times 2=14$
$\rightarrow 6+5=11\times 4=44$ Hence, the answer is 49.
$\rightarrow 5+2=7\times 7=49$

Q86. Solution

Correct Answer: (C)

Given series is: $2, 12, 36, 80, 150$
$2 \rightarrow (1)+(1)^3$
$12 \rightarrow (2)^2+(2)^3$
$36 \rightarrow (3)^2+(3)^3$
$80 \rightarrow (4)^2+(4)^3$
$150 \rightarrow (5)^2+(5)^3$
$? \rightarrow (6)^2+(6)^3=36+216$
$=252$

Q87. Solution

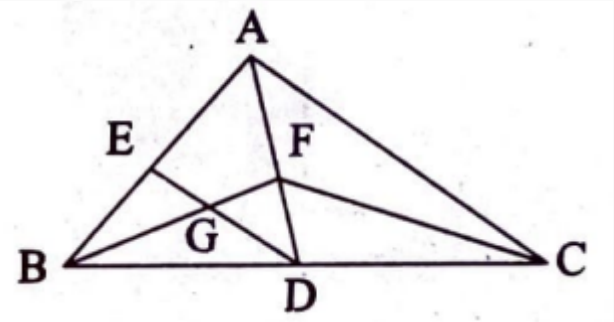
Correct Answer: (B)

$$\begin{aligned} &\text{When} \quad \rightarrow \times \\ &\text{You} \quad \rightarrow \div \\ &\text{Come} \quad \rightarrow - \\ &\text{Will} \quad \rightarrow + \end{aligned}$$

8 When 12 Will 16 You 2 Come 10 $=?$
Using the correct symbols, we have
$=8\times 12+16\div 2-10$
$=96+8-10$
$=104-10=94$

Q88. Solution

Correct Answer: (B)



The triangles are:
$\triangle$ABC; $\triangle$ABD; $\triangle$ADC; $\triangle$AFC ΔFDC; ΔAFB; ΔFDB; ΔFBC ΔGBD; ΔADE; ΔGBE; ΔFDG $\triangle$DBE

Q89. Solution

Correct Answer: (D)

Bonus and dividend are different from each other but both these are part of profit.

Q90. Solution

Correct Answer: (A)

Figure given in option (a) can replace the question mark.

Q91. Solution

Correct Answer: (B)

We have $\sec^2\theta = \frac{4}{3} \Rightarrow \cos^2\theta = \frac{3}{4}$
$\Rightarrow \cos^2\theta = \cos^2\left(\frac{\pi}{6}\right) \Rightarrow$
$\theta = n\pi \pm \frac{\pi}{6} \ldots\ldots\ldots$

Q92. Solution

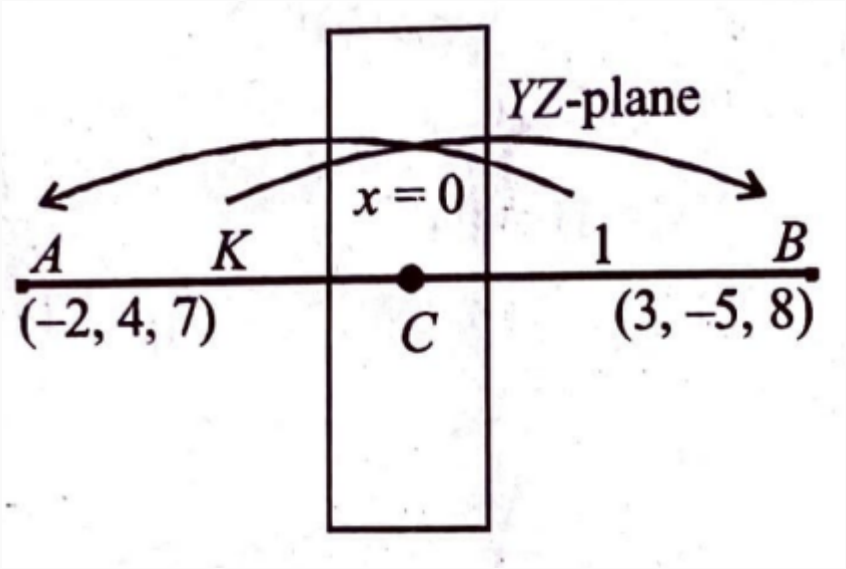
Correct Answer: (B)

There are 6 letters in the word BHARAT, 2 of them are identical. Hence total number of words $= 6!/2! = 360$
Number of words in which B and H come together
$= \frac{5!2!}{2!} = 120$
final ans $= 360 - 120 = 240$

Q93. Solution

Correct Answer: (B)

Let the points be $A(-2,4,7)$ and $B(3,-5,8)$ on YZ-plane, x-coordinate $=0$.



Let the ratio be $K:1$.

The coordinates of C are $\left(\frac{3K-2}{K+1}, \frac{-5K+4}{K+1}, \frac{8K+7}{K+1}\right)$

Clearly $\frac{3K-2}{K+1}=0 \Rightarrow 3K=2 \Rightarrow K=\frac{2}{3}$

Hence required ratio is $2:3$.

Q94. Solution

Correct Answer: (C)

We have

$\min n(A \cup B)=\max\{n(A), n(B)\}=\max\{3,6\}=6$

$\text{Max}\, n(A \cup B)=n(A)+n(B)=9$

$\therefore 6 \leq n(A \cup B) \leq 9$

Q95. Solution

Correct Answer: (D)

Let the statement $\mathrm{P(n)}$ be defined as $P(n): \frac{n^5}{5}+\frac{n^3}{3}+\frac{7n}{15}$ is a natural number for all $n \in N$:

Step I : For $\mathrm{n}=1, \mathrm{P}(1): \frac{1}{5}+\frac{1}{3}+\frac{7}{15}=1 \in \mathrm{N}$

Hence, it is true for $\mathrm{n}=1$.

Step II : Let it is true for $n=k$,

i.e. $\frac{k^5}{5}+\frac{k^3}{3}+\frac{7k}{15}=\lambda \in N \ldots$ (i)

Step III : For $\mathrm{n}=\mathrm{k}+1$,

$\frac{(\mathrm{k}+1)^5}{5}+\frac{(\mathrm{k}+1)^3}{3}+\frac{7(\mathrm{k}+1)}{15}$

$=\frac{1}{5}\left(k^5+5k^4+10k^3+10k^2+3k+1\right)+\frac{1}{3}\left(k^3+3k^2+3k+1\right)+\frac{7}{15}k+\frac{7}{15}$

$=\left(\frac{k^5}{5}+\frac{k^3}{3}+\frac{7}{15}k\right)+\left(k^4+2k^3+3k^2+2k\right)+\frac{1}{5}+\frac{1}{3}+\frac{7}{15}$

$=\lambda+\mathrm{k}^4+2\mathrm{k}^3+3\mathrm{k}^2+2\mathrm{k}+1$ [using equation (i)]

which is a natural number, since $\lambda \mathrm{k} \in \mathrm{N}$. Therefore, $P(k+1)$ is true, when $P(k)$ is true. Hence, from the principle of mathematical induction, the statement is true for all natural numbers n

Q96. Solution

Correct Answer: (C)

Given equation is

$(p-q)x^2+(q-r)x+(r-p)=0$

By using formula for finding the root viz: $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$, we get

$x=\frac{(r-q) \pm \sqrt{(q-r)^2-4(r-p)(p-q)}}{2(p-q)}$

$\Rightarrow \quad x=\frac{(r-q) \pm(q+r-2p)}{2(p-q)}=\frac{r-p}{p-q}, 1$

Q97. Solution

Correct Answer: (A)

The given lines are:

$y=(2-\sqrt{3})x+5$ and $y=(2+\sqrt{3})x-7$

Therefore, slope of first line $=m_1=2-\sqrt{3}$ and slope of second line $=m_2=2+\sqrt{3}$

$\therefore \quad \tan \theta=\frac{m_2-m_1}{1+m_1 m_2}=\frac{2+\sqrt{3}-2+\sqrt{3}}{1+(4-3)}$

$=\frac{2\sqrt{3}}{2}=\sqrt{3}=\tan \frac{\pi}{3} \Rightarrow \theta=\frac{\pi}{3}=60^\circ$

Q98. Solution

Correct Answer: (C)

$f(x)$ is defined if $3x^2-4x+5 \geq 0$

$\Rightarrow 3\left[x^2-\frac{4}{3}x+\frac{5}{3}\right] \geq 0 \Rightarrow 3\left[\left(x-\frac{2}{3}\right)^2+\frac{11}{9}\right] \geq 0$

Which is true for all real $x \therefore$ Domain of $f(x)=(-\infty, \infty)$

Let $y=\sqrt{3x^2-4x+5}$

$\Rightarrow y^2=3x^2-4x+5$ i.e. $3x^2-4x+\left(5-y^2\right)=0$

For x to be real , $16-12\left(5-y^2\right) \geq 0 \Rightarrow y \geq \sqrt{\frac{11}{3}}$

$\therefore$ Range of $y=\left[\sqrt{\frac{11}{3}}, \infty\right)$

Q99. Solution

Correct Answer: (B)

$f(x)=\frac{x}{\sqrt{1+x^2}}$

fof$=\frac{\frac{x}{\sqrt{1+x^2}}}{\sqrt{1+\frac{x^2}{1+x^2}}}=\frac{x}{\sqrt{2x^2+1}}$

fofof $=\frac{\frac{x}{\sqrt{2x^2+1}}}{\sqrt{1+\frac{x^2}{2x^2+1}}}=\frac{x}{\sqrt{1+3x^2}}$

Q100. Solution

Correct Answer: (C)

Let $y=\mathrm{e}^{\mathrm{x}^3}, \mathrm{z}=\log \mathrm{x}$

On differentiating w.r.t.x, we get

$\frac{dy}{dx}=e^{x^3}\left(3x^2\right)=3x^2e^{x^3}$ and $\frac{dz}{dx}=\frac{1}{x}$

$\therefore \frac{dy}{dz}=\frac{\frac{dy}{dx}}{\frac{dz}{dx}}=\frac{3x^2e^{x^3}}{\left(\frac{1}{x}\right)}=3x^3e^{x^3}$

Q101. Solution

Correct Answer: (A)

Equation of line AB is $y-3=\frac{6-3}{7-3}(x-3)$

$\Rightarrow 3x-4y+3=0 \Rightarrow \frac{x}{-1}+\frac{y}{3/4}=1$

Hence required length is $\sqrt{(-1)^2+\left(\frac{3}{4}\right)^2}=\frac{5}{4}$.

Q102. Solution

Correct Answer: (D)

$2^x + 2^{|x|} \geq 2\sqrt{2}$ ...(i)

Case I: $x \geq 0$, then Eq. (i) becomes

$2^x + 2^x \geq 2\sqrt{2}$

$\Rightarrow 2^x \geq \sqrt{2} \Rightarrow x \geq \frac{1}{2}$

Case II: $x < 0$, then Eq. (i) becomes

$2^x + 2^{-x} \geq 2\sqrt{2}$

$\Rightarrow t + \frac{1}{t} \geq 2\sqrt{2}$, where $2^x = t$

$\Rightarrow t^2 - 2\sqrt{2}t + 1 \geq 0$

$\Rightarrow x \leq \log_2(\sqrt{2} - 1)$

Also, $0 < \sqrt{2} - 1 < 1, \log_2(\sqrt{2} - 1) < 0$.

$\therefore$ The solution is

$\left(-\infty, \log_2(\sqrt{2} - 1)\right)\Big] \cup \left[\frac{1}{2}, \infty\right)$

Q103. Solution

Correct Answer: (A)

$y = \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}$

$\Rightarrow y = \sqrt{\frac{2\cos^2\theta}{2\sin^2\theta}} = \sqrt{\cot^2\theta}$

$\Rightarrow y = \cot\theta$

Differentiate w.r.t. ' θ ', we get: $\frac{dy}{d\theta} = -\operatorname{cosec}^2\theta$

Now, $\left(\frac{\mathrm{dy}}{\mathrm{d}\theta}\right)_{\theta=\frac{3\pi}{4}} = -\operatorname{cosec}^2\left(\frac{3\pi}{4}\right)$

$= -\operatorname{cosec}^2\left(\pi - \frac{\pi}{4}\right) = -\operatorname{cosec}^2\frac{\pi}{4} = -2$

Q104. Solution

Correct Answer: (B)

Since, $\frac{dy}{dx} = \frac{y+1}{x-1}$

$\Rightarrow \quad \frac{dy}{y+1} = \frac{dx}{x-1}$

After integrating on both sides, we have

$\log(y+1) = \log(x-1) - \log C$

$C(y+1) = (x-1)$

$C = \frac{x-1}{y+1}$

If $x = 1$, then $y = 2$, so $C = 0$

Therefore, $x - 1 = 0$

Hence, there is only one solution.

Q105. Solution

Correct Answer: (B)

Here $n(S) = 6^2 = 36$

Let E be the event "getting sum more than 7" i.e. sum of pair of dice $= 8, 9, 10, 11, 12$

i.e. $E = \left\{ \begin{matrix} (2,6) & (3,5) & (4,4) & (5,3) & (6,2) \\ (3,6) & (4,5) & (5,4) & (6,3) & \\ (4,6) & (5,5) & (6,4) & & \\ (5,6) & (6,5) & (6,6) & & \end{matrix} \right\}$

$\therefore \quad n(E) = 15. \therefore$ Req. prob $= \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$

Q106. Solution

Correct Answer: (C)

Total no. of cards $= 52$

13 cards are diamonds and 4 cards are king.

There is only one card which is a king of diamond.

$\therefore$ P(card is diamond) $= \frac{13}{52}$, P (card is king)

$= \frac{4}{52}$

P(card is king of diamond) $= \frac{1}{52}$

$\therefore \quad P$ (card is diamond or king)

$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

Q107. Solution

Correct Answer: (B)

$kA = \begin{bmatrix} 0 & 2k \\ 3k & -4k \end{bmatrix} = \begin{bmatrix} 0 & 3a \\ 2b & 24 \end{bmatrix}$

$\Rightarrow \quad k = -6, a = -4$ and $b = -9$

Q108. Solution

Correct Answer: (C)

formula used $L = \dfrac{2b^2}{a}$, $e = \sqrt{1} - \dfrac{b^2}{a^2}$ Length of transverse axis $= 2\mathrm{a}$ Given

$$\mathrm{e} = \frac{\sqrt{13}}{3},\ \mathrm{L} = \frac{10}{3}$$
$$\frac{2\,\mathrm{b}^2}{\mathrm{a}} = \frac{10}{3}$$
$$\Rightarrow\ \mathrm{b}^2 = \frac{5\mathrm{a}}{3}$$
$$\mathrm{b}^2 = \mathrm{a}^2\left(\mathrm{e}^2 - 1\right)$$
$$\Rightarrow \frac{5\mathrm{a}}{3} = \mathrm{a}^2\left(\frac{13}{9} - 1\right)$$
$$\Rightarrow \frac{5\mathrm{a}}{3} = \mathrm{a}^2\left(\frac{4}{9}\right)$$
$$\Rightarrow 4\mathrm{a}^2 - 15\mathrm{a} = 0$$
$$\Rightarrow \mathrm{a}(4\mathrm{a} - 15) = 0$$
$$\mathrm{a} = 0, \frac{15}{4}$$
$$\mathrm{a} \neq 0$$

Hence $\mathrm{a} = \frac{15}{4}$

Length of transverse axis $= 2\mathrm{a}$
$$= \frac{15}{4}(2)$$
$$= \frac{15}{2}$$

Q109. Solution

Correct Answer: (B)

Sum of infinite terms of series a $+ar + ar^2 + = 15$

$\therefore \frac{a}{1-r} = 15$ ...(i)

Sequence formed by square of terms: $a^2, a^2r^2, a^2r^4, a^2r^6 \ldots$

$a^2 + a^2r^2 + \cdots = 150$

Sum $= \frac{a^2}{1-r^2} = 150$

$\Rightarrow \frac{a}{1-r} \cdot \frac{a}{1+r} = 15 \Rightarrow 15 \cdot \frac{a}{1+r} = 150\ \left(\because \frac{\mathrm{a}}{1-\mathrm{r}} = 15\right)$

$\Rightarrow \frac{a}{1+r} = 10$ ...(ii)

On dividing equation (i) by (ii)

$\frac{1+\mathrm{r}}{1-\mathrm{r}} = \frac{15}{10}$

or $r = \frac{1}{5} \Rightarrow a = 12$

Now series : $ar^2 + ar^4 + ar^6 + \ldots$

Sum $= \frac{a^2}{1-r^2} = \frac{12\cdot\left(\frac{1}{25}\right)}{1-\frac{1}{25}} = \frac{1}{2}$

Q110. Solution

Correct Answer: (A)

Given $f(x)=\frac{4x^2+1}{x}$ Thus $f'(x)=4-\frac{1}{x^2}$

$\mathrm{f(x)}$ will be decreasing if $\mathrm{f'(x)<0}$

Thus $4-\frac{1}{x^2}<0 \Rightarrow \frac{1}{x^2}>4 \Rightarrow \frac{-1}{2}<x<\frac{1}{2}$

Thus interval in which $f(x)$ is decreasing, is $\left(-\frac{1}{2}, \frac{1}{2}\right)$.

Q111. Solution

Correct Answer: (C)

$\int e^x \frac{(1+\sin x)}{(1+\cos x)} dx$

$=\int e^x\left[\frac{1}{2}\sec^2\frac{x}{2}+\tan\frac{x}{2}\right]dx$

$=\frac{1}{2}\int e^x \sec^2\frac{x}{2}dx+\int e^x\tan\frac{x}{2}dx$

$=\mathrm{e^x}\tan\frac{\mathrm{x}}{2}+\mathrm{C}$

But $\mathrm{I=e^x f(x)+C}$ (given)

$\therefore \mathrm{f(x)=\tan\frac{x}{2}}$

Q112. Solution

Correct Answer: (B)

$\because x+y=e^{xy}$

Differentiating w.r.t. x, we get

$1+\frac{dy}{dx}=e^{xy}\left[y+x\frac{dy}{dx}\right]$

$\Rightarrow \quad \frac{dy}{dx}(1-xe^{xy})=ye^{xy}-1$

$\Rightarrow \frac{dy}{dx}=\frac{ye^{xy}-1}{1-xe^{xy}} \because \frac{dy}{dx}=\infty,$

as tangent is parallel to Y-axis

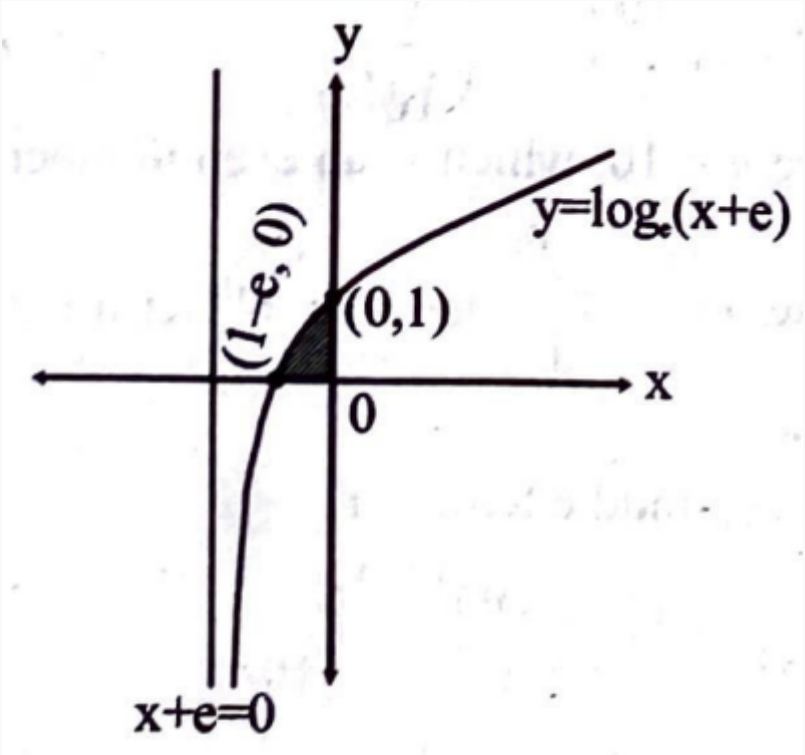
$\Rightarrow 1-xe^{xy}=0$

$\therefore \mathrm{xe^{xy}}=1$

This holds, when $x=1$ and $y=0$

Q113. Solution

Correct Answer: (A)



Required area $A = \int_{1-e}^{0} y dx = \int_{1-e}^{0} \log_e(x+e) dx$

put $x + e = t \Rightarrow dx = dt$ also when $x = 1 - e, t = 1$ and when $x = 0, t = e$

$\therefore A = \int_1^e \log_e t dt = [t \log_e t - t]_1^e$

$e - e - 0 + 1 = 1$

Q114. Solution

Correct Answer: (D)

Given $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{a} \cdot \vec{b} = 1$ and $\vec{a} \times \vec{b} = \vec{j} - \vec{k}$,

Let $\bar{b} = b_1\hat{i} + b_2\hat{j} + b_3\widehat{k}$

Now, $\hat{j} - \widehat{k} = \bar{a} \times \bar{b} = \begin{matrix} \hat{i} & \hat{j} & \widehat{k} \\ 1 & 1 & 1 \\ b_2 & b_2 & b_3 \end{matrix}$

$\Rightarrow b_3 - b_2 = 0, b_1 - b_3 = 1, b_2 - b_1 = -1$

$\Rightarrow b_3 = b_2, b_1 = b_2 + 1$

Also, $\bar{a} \cdot \bar{b} = 1$

$\Rightarrow b_1 + b_2 + b_3 = 1$

$\Rightarrow 3b_2 + 1 = 1$

$\Rightarrow b_2 = 0$

$\Rightarrow b_1 = 1, b_3 = 0$

Thus, $\bar{b} = \hat{i}$

Q115. Solution

Correct Answer: (C)

Let $l=\lim_{x \rightarrow 0} \frac{[\sin x]}{x}$; RHL $=\lim_{x \rightarrow 0^{+}} \frac{\sin x}{x}=1$

LHL $=\lim_{x \rightarrow 0^{-}}\left(\frac{-\sin x}{x}\right)=-\left(\lim_{x \rightarrow 0^{-}} \frac{\sin x}{x}\right)=-1$

$\left[|x|=\begin{array}{l} x \text{ if } x \geq 0 \\ -x \text{ if } x \leq 0 \end{array}\right]$

As, LHL $\neq$ RHL Hence, limit l.does not exist.

Q116. Solution

Correct Answer: (D)

Two planes are coplanar if

$$\begin{array}{ccc} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{array}=0$$

$$\begin{array}{ccc} 1 & -1 & -1 \\ 1 & 1 & -k \\ k & 2 & 1 \end{array}=0$$

Applying $\mathrm{C}_2 \rightarrow \mathrm{C}_2+\mathrm{C}_1, \mathrm{C}_3 \rightarrow \mathrm{C}_3+\mathrm{C}_1$

$$\Rightarrow \begin{array}{ccc} 1 & 0 & 0 \\ 1 & 2 & 1-k \\ k & k+2 & 1+k \end{array}=0$$

$\Rightarrow \quad 1[2+2k-(k+2)(1-k)]=0$

$\Rightarrow \quad 2+2k-\left(-k^2-k+2\right)=0$

$k^2+3k=0 \Rightarrow k(k+3)=0$

Q117. Solution

Correct Answer: (D)

Given expression is $p \Leftrightarrow(q \Rightarrow p)$

$\sim(\mathrm{p} \leftrightarrow(\mathrm{q} \rightarrow \mathrm{p}))$

$\sim(\mathrm{p} \leftrightarrow \mathrm{q})=(\mathrm{p} \wedge \sim \mathrm{q}) \vee(\mathrm{q} \wedge \sim \mathrm{p})$

$\sim(\mathrm{p} \leftrightarrow(\mathrm{q} \rightarrow \mathrm{p}))=(\mathrm{p} \wedge \sim(\mathrm{q} \rightarrow \mathrm{p})) \vee((\mathrm{q} \rightarrow \mathrm{p}) \wedge \sim \mathrm{p})$

$(\mathrm{p} \wedge \sim(\mathrm{q} \rightarrow \mathrm{p}))=\mathrm{p} \wedge(\mathrm{q} \wedge \sim \mathrm{p})=(p \wedge \sim p) \wedge q=c$

$(q \rightarrow p) \wedge \sim p=(\sim q \vee p) \wedge \sim p=\sim p \wedge(\sim q \vee p)$

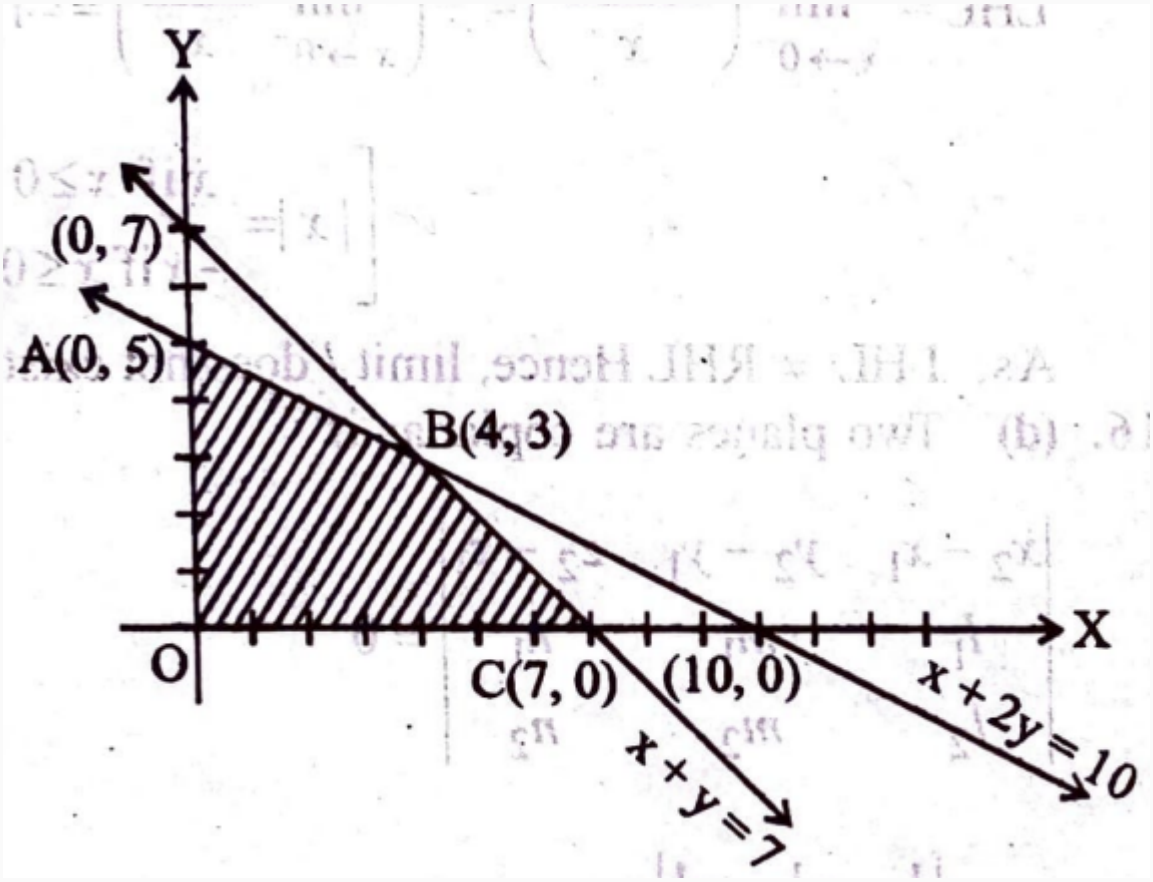
$=(\sim \mathrm{p} \wedge \sim \mathrm{q}) \vee(\sim \mathrm{p} \wedge \mathrm{p})=\sim \mathrm{p} \wedge \sim \mathrm{q}$

$\sim(p \leftrightarrow(\dot{q} \rightarrow p))=c \vee(\sim p \wedge \sim q)=\sim p \wedge \sim q$

Q118. Solution

Correct Answer: (C)

Change the inequalities into equations and draw the graph of lines, thus we get the required feasible region as shown below.



The region bounded by the vertices $A(0,5), B(4,3), C(7,0)$

The objective function is maximum at $\mathrm{C}(7,0)$ and Max $z = 5 \times 7 + 2 \times 0 = 35$.

Q119. Solution

Correct Answer: (A)

Arithmetic mean $\bar{x}$ of $4, 7, 8, 9, 10, 12, 13, 17$ is

$\bar{x} = \frac{4+7+8+9+10+12+13+17}{8} = \frac{80}{8} = 10$

$\Sigma\,|x_i - \bar{x}| = 6 + 3 + 2 + 1 + 0 + 2 + 3 + 7 = 24$

$\therefore$ Mean deviation about mean

$= \text{ M.D. } (\bar{x}) = \frac{\sum|x_i - \bar{x}|}{n} = \frac{24}{8} = 3$

Q120. Solution

Correct Answer: (B)

Let E_1, E_2 and A be the events defined as follows:

$\mathrm{E}_1 =$ red ball is transferred from bag P to bag Q

$\mathrm{E}_2 =$ blue ball is transferred from bag P to bag Q

$\mathrm{A} =$ the ball drawn from bag Q is blue

As the bag P contains 6 red and 4 blue balls,

$P(E_1) = \frac{6}{10} = \frac{3}{5}$ and $P(E_2) = \frac{4}{10} = \frac{2}{5}$

Note that E_1 and E_2 are mutually exclusive and exhaustive events.

When E_1 has occurred i.e., a red ball has already been transferred from bag P to Q, then bag Q will contain 6 red and 6 blue balls, So, $P(A \mid E_1) = \frac{6}{12} = \frac{1}{2}$

When E_2 has occurred i.e., a blue ball has already been transferred from bag P to Q, then bag Q will contain 5 red and 7 blue balls, So, $P(A \mid E_2) = \frac{7}{12}$

By using law of total probability, we get

$P(A) = P(E_1)P(A \mid E_1) + P(E_2)P(A \mid E_2)$

$= \frac{3}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{7}{12} = \frac{8}{15}$

Q121. Solution

Correct Answer: (D)

$\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) = \tan^{-1}\left\{\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right\}$

$= \tan^{-1}\left\{\frac{9+8}{36-2}\right\} = \tan^{-1}\left(\frac{1}{2}\right)$

Q122. Solution

Correct Answer: (A)

General term $= T_{r+1}$

$= {}^{10}\mathrm{C}_r\left(\frac{10}{x}\right)^{10-r}\left(\frac{x}{10}\right)^r$

Here $n = 10$, which is an even number.

Now, $\left[\frac{10}{2}+1\right]^{\mathrm{th}}$ term i.e. 6th term is the middle term.

Hence, middle term $= T_6$

$\mathrm{T}_{5+1} = {}^{10}\mathrm{C}_5\left(\frac{10}{x}\right)^{10-5}\left(\frac{x}{10}\right)^5$

$= {}^{10}\mathrm{C}_5\left(\frac{10}{x}\right)^5\left(\frac{x}{10}\right)^5 = {}^{10}\mathrm{C}_5.$

Q123. Solution

Correct Answer: (B)

Equation of tangent of parabola $y = x^2$ be

$tx = y + at^2$ ...(i)

$y = tx - \frac{t^2}{4}$

Solve with $\mathrm{y} = -(\mathrm{x}-2)^2$

$tx - \frac{t^2}{4} = -(x-2)^2$

$x^2 + x(t-4) - \frac{t^2}{4} + 4 = 0$

Here, Discriminant $= 0$.

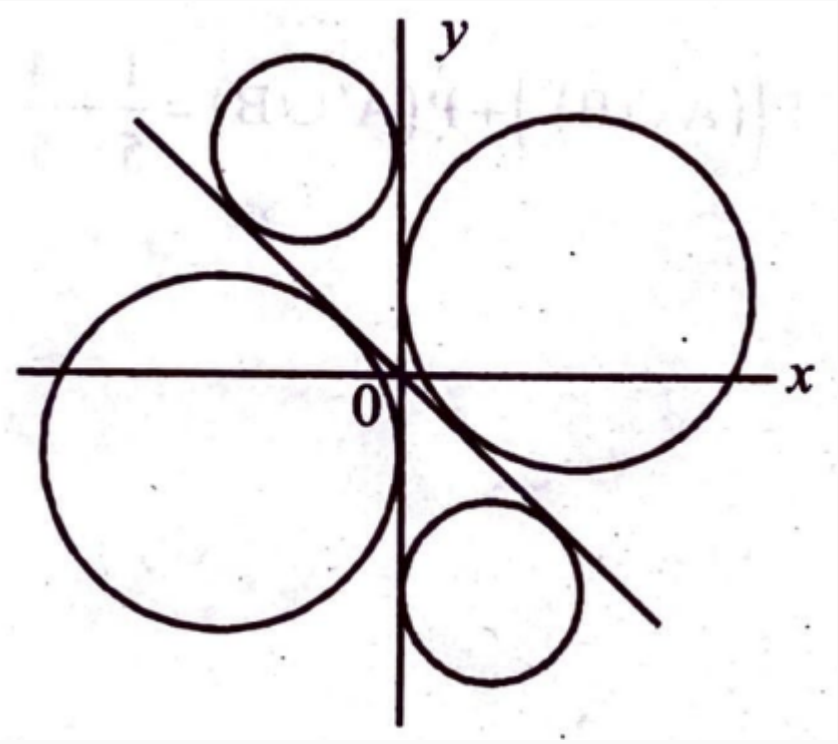
$(t-4)^2 - 4 \cdot \left(4 - \frac{t^2}{4}\right) = 0 \Rightarrow t^2 - 4t = 0 \Rightarrow t = 0$

or $\mathrm{t} = 4$

Put value of t in eq. (i), then $y = 4(x-1)$.

Q124. Solution

Correct Answer: (D)



Let (h, k) is centre of circle

$\frac{h-k}{\sqrt{2}} = |h|, k^2 - h^2 + 2hk = 0$

Q125. Solution

Correct Answer: (B)

$\because \mathrm{f(x)} = \tan^{-1}(\sin \mathrm{x} + \cos \mathrm{x}) \therefore f'(x) = \frac{\cos x - \sin x}{1+(\sin x + \cos x)^2} = \frac{\sqrt{2}\cos\left(x+\frac{\pi}{4}\right)}{1+(\sin x+\cos x)^2}$ $\mathrm{f(x)}$ is increasing if $-\frac{\pi}{2} < \mathrm{x} + \frac{\pi}{4} < \frac{\pi}{2} \Rightarrow -\frac{3\pi}{4} < \mathrm{x} < \frac{\pi}{4}$ Hence, $f(x)$ is increasing when $x \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)$

Q126. Solution

Correct Answer: (A)

$i^{57}+\frac{1}{i^{25}}=\left(i^{4}\right)^{14}.i+\frac{1}{(i^{4})^{6}.i}=i+\frac{1}{i}=i-i=0$

Q127. Solution

Correct Answer: (D)

4 is a root of $x^2+px+12=0$

$\Rightarrow 16+4p+12=0 \Rightarrow p=-7$

Now, the equation $x^2+px+q=0$ has equal roots.

$\therefore p^2-4q=0 \Rightarrow q=\frac{p^2}{4}=\frac{49}{4}$

Q128. Solution

Correct Answer: (A)

Suppose,

$I=\int \frac{x+3}{(x+4)^2}e^x dx=\int \frac{(x+4)1}{(x+4)^2}e^x dx$

$=\int \frac{e^x}{(x+4)}-\int \frac{e^x}{(x+4)^2}dx$

$=\int e^x\left(\frac{1}{(x+4)}-\frac{1}{(x+4)^2}\right)dx$

$=e^x\frac{1}{(x+4)}+C$

$\left[\int e^x\{f(x)+f'(x)\}dx=e^x f(x)+C\right]$

Q129. Solution

Correct Answer: (A)

Give lines are $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-1}{-1}$ and $\frac{x+3}{2}=\frac{y-6}{1}=\frac{z-5}{3}$

Lines passes through the points

$\vec{a}_1=(3,2,1)$ and $\vec{a}_2=(-3,6,5)$,

$\vec{b}_1=2\hat{i}+3\hat{j}-\hat{k}$

$\vec{b}_1=2\hat{i}+\hat{j}-3\hat{k}, \vec{a}_2-\vec{a}_1=6\hat{i}-4j-4\hat{k}$

Shortest distance $=\frac{\left(\vec{a}_2-\vec{a}_1\right)\left(\vec{b}_1\times\vec{b}_2\right)}{\left(\vec{b}_1\times\vec{b}_2\right)}$

$$\vec{b}_1\times\vec{b}_2=\begin{matrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & -1\\ 2 & 1 & 3\end{matrix}=10\hat{i}-8\hat{j}-4\hat{k}$$

$\left(\vec{a}_2-\vec{a}_1\right)\cdot\left(\vec{b}_1\times\vec{b}_2\right)=60+32+16=108$

$\vec{b}_1\times\vec{b}_2=\sqrt{100+64+16}=\sqrt{180}$

$S.D=\frac{108}{\sqrt{180}}=\frac{108}{6\sqrt{5}}=\frac{18}{\sqrt{5}}$

Q130. Solution

Correct Answer: (D)

$P(B)=\frac{3}{5}, P(A \mid B)=\frac{1}{2}$ and $P(A \cup B)=\frac{4}{5}$

$P(A \cap B)=P(A \mid B) P(B)=\frac{1}{2} \cdot \frac{3}{5}=\frac{3}{10}$

$\mathrm{P}(\mathrm{A} \cup \mathrm{B})=\mathrm{P}(\mathrm{A})+\mathrm{P}(\mathrm{B})-\mathrm{P}(\mathrm{A} \cap \mathrm{B})$

$\mathrm{P}(\mathrm{A})=\frac{4}{5}-\frac{3}{10}=\frac{1}{2}. \quad \mathrm{P}\left(\mathrm{A}^{\prime}\right)=1-\mathrm{P}(\mathrm{A})=\frac{1}{2}$

We know, $\mathrm{P}(\mathrm{A} \cap \mathrm{B})+\mathrm{P}\left(\mathrm{A}^{\prime} \cap \mathrm{B}\right)=\mathrm{P}(\mathrm{B})$

[as $A \cap B$ and $A' \cap B$ are mutually exclusive events]

$\Rightarrow \quad \frac{3}{10}+\mathrm{P}\left(\mathrm{A}^{\prime} \cap \mathrm{B}\right)=\frac{3}{5}$

$\Rightarrow \mathrm{P}\left(\mathrm{A}^{\prime} \cap \mathrm{B}\right)=\frac{3}{5}-\frac{3}{10}=\frac{3}{10}$

Now, $\mathrm{P}\left(\mathrm{A}^{\prime} \cup \mathrm{B}\right)=\mathrm{P}\left(\mathrm{A}^{\prime}\right)+\mathrm{P}(\mathrm{B})-\mathrm{P}\left(\mathrm{A}^{\prime} \cap \mathrm{B}\right)$

$=\frac{1}{2}+\frac{3}{5}-\frac{3}{10}=\frac{5+6-3}{10}=\frac{4}{5}$

$\mathrm{P}\left((\mathrm{A} \cup \mathrm{B})^{\prime}\right)=1-\mathrm{P}(\mathrm{A} \cup \mathrm{B})=1-\frac{4}{5}=\frac{1}{5}$

$\therefore \mathrm{P}\left((\mathrm{A} \cup \mathrm{B})^{\prime}\right)+\mathrm{P}\left(\mathrm{A}^{\prime} \cup \mathrm{B}\right)=\frac{1}{5}+\frac{4}{5}=1$

Q131. Solution

Correct Answer: (A)

We have, angular velocity is differentiation of angular position and angular acceleration is derivative of angular velocity.

Torque (T) is cross product of force (F) and distance (r) which also equal to Moment of Inertia (I) product by angular acceleration (α).

Torque $= r \times F = I\alpha$

So, $2\left(20t - 5t^2\right) = 10\alpha$

$\alpha = 4t - t^2$

But $\alpha = \frac{d\omega}{dt}$, so

$\frac{d\omega}{dt} = 4t - t^2$

$d\omega = \left(4t - t^2\right)dt$

Integrating, $\omega = 2t^2 - \frac{t^3}{3}$

ω will be zero at $t = 6$ s.

So, $\omega = \frac{d\theta}{dt} = 2t^2 - \frac{t^3}{3}$

or $d\theta = \left(2t^2 - \frac{t^3}{3}\right)dt$

Again, integrating both sides, we get

$\theta = \frac{2t^3}{3} - \frac{t^4}{12}$

Since, $t = 6$ s

so, $\theta = \frac{2\times 6^3}{3} - \frac{6^4}{12} = 36$

Number of turns, $n = \frac{\theta}{2\pi} = \frac{36}{2\pi} = 5.73$

Q132. Solution

Correct Answer: (A)

Given, magnetic field

$$\overrightarrow{B} = B\left(\cos 45^{o}\hat{i} + \sin 45^{o}\hat{j}\right) = \frac{B}{\sqrt{2}}\left(\hat{i} + \hat{j}\right)$$

$$\overrightarrow{A} = L^{2}\widehat{k}$$

$$\overrightarrow{\tau} = \overrightarrow{M} \times \overrightarrow{B} = i\left(\overrightarrow{A} \times \overrightarrow{B}\right) = i_{0}\left[L^{2}\widehat{k} \times \frac{B}{\sqrt{2}}\left(\hat{i} + \hat{j}\right)\right] = \frac{i_{0}L^{2}B}{\sqrt{2}}\left(-\hat{i} + \hat{j}\right)$$

It is clear from above result that axis of rotation is$= \left(\frac{-\hat{i}+\hat{j}}{\sqrt{2}}\right)$i.e. SQ

Moment of inertia of the frame about axis of rotation SQ.

Moment of inertia of the frame about an axis passing through O and perpendicular to the plane of the frame is given.

$$I_{0} = \frac{4}{3}ML^{2}$$

By perpendicular axis theorem

$$2I_{SQ} = I$$

$$\therefore \qquad I_{SQ} = \frac{I_{0}}{23} = \frac{2}{3}ML^{2}$$

$$\text{We know that} \quad \tau = I\alpha$$

$$\therefore \qquad \alpha = \frac{\tau}{I} = \frac{i_{0}BL^{2}}{\frac{2}{3}ML^{2}} = \frac{3}{2}\ \frac{i_{0}B}{M}$$

Thus angular rotation

$$\theta = \frac{1}{2I}\alpha(\Delta t)^{2} = \frac{1}{2}\left(\frac{3i_{0}B}{2M}\right)\Delta t^{2} = \frac{3}{4}\ \frac{i_{0}B}{M}\Delta t^{2}$$

Q133. Solution
Correct Answer: (B)

Given, optical path difference, $\Delta x = 260\frac{\lambda}{4} = 65\lambda$.

According to Young's double-slit experiment, the condition for bright fringe (maximum intensity) is given as, $\Delta x = n\lambda$. Here, $n = 0, 1, 2, ...$ is an integer.

The condition for dark fringe (minimum intensity) is given as, $\Delta x = (2n-1)\frac{\lambda}{2}$.

From the above two conditions, on comparing it with the given path difference, for maximum intensity, $n = 65$. Thus, the point will be bright. Since the value of n is an integer and in the case of dark fringe, the value of n will be decimal for given path difference, thus, it is a bright fringe.

Q134. Solution
Correct Answer: (A)

When amino acids with free alpha-amino groups are treated with an excess of ninhydrin, they yield a purple coloured product. The colour intensity depends on the concentration of amino acid.

Ninhydrin, which is originally yellow, when reacts with amino acid, it turns deep purple. It reacts with the free amino group. This group is present in all the amino acids, proteins or peptides. Theoretically, only amino acids produce colour with the ninhydrin reagent.

Q135. Solution
Correct Answer: (C)

$CrCl_3 \cdot 6H_2O$ has $3-\mathrm{Cl}$ atoms but $\frac{1}{3}$ Cl atoms precipitated so one $1\,\mathrm{Cl}$ atom get precipitated.

According to Werner's theory, only those ions are precipitated which are attached to the metal atoms with ionic bonds and are present outside the co-ordination sphere.

$$[Cr(H_2O)_4Cl_2]Cl\,(aq) \rightleftharpoons \quad [Cr(H_2O)_4Cl_2]^+(aq) + Cl^-(aq)$$

$$Ag^+ + Cl^- \longrightarrow AgCl \downarrow (\text{white})$$

Q136. Solution
Correct Answer: (A)

Any redox reaction would occur spontaneously if the free energy change (ΔG) is negative. $\Delta G^o = -nFE^o$ Where n is the number of electrons involved, F is the value of Faraday and E^o is the cell emf. ΔG^o can be negative if E^o is positive.

Q137. Solution
Correct Answer: (C)

The given logic is,

$(352) + (422)$
$3 + 5 + 2 + 4 + 2 + 2 = 18$
$(712) + (735)$
$7 + 1 + 2 + 7 + 3 + 5 = 25$
$(452) + (328)$
$4 + 5 + 2 + 3 + 2 + 8 = 24$

Hence, correct answer is 24.

Q138. Solution
Correct Answer: (C)

According to the question,

An idiot means stupid or foolish which is deficient in intelligence. Similarly, a desert is a dry place which is deficient in water.

Erudite means having or showing great knowledge or a learning.

Wealthy is having a great deal of money, resources, or assets, rich.

Lackadaisical means lacking an enthusiasm and a determination.

Hence, the correct answer is Desert.

Q139. Solution
Correct Answer: (C)

In the first pair, the numbers are related as

$(10)^2 = 100$

We see that squaring the first number gives second number of the same pair.

Similarly, we can solve for the second pair in following manner,

$(14)^2 = 196.$

Q140. Solution

Correct Answer: (C)

We know that $A \vee B$ is false only when both A & B are false and $A \wedge B$ is true only when both A & B are true.

Let, $s : p \wedge [(\sim q \vee r) \wedge (\sim r \vee q)]$

p	q	r	$\sim q$	$\sim r$	$\sim q \vee r$	$\sim r \vee q$	$(\sim q \vee r) \wedge (\sim r \vee q)$	s
F	F	F	T	T	T	T	T	F

As per the truth table given above, it is clear that s has truth value F and only option $(\sim q \vee r) \wedge (\sim r \vee q)$ is false since $(\sim q \vee r) \wedge (\sim r \vee q)$ T is true T.

Q141. Solution

Correct Answer: (B)

The volume of the parallelepiped is, $v = \left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$$\Rightarrow \begin{vmatrix} 1 & 1 & n \\ 2 & 4 & -n \\ 1 & n & 3 \end{vmatrix} = \pm 158$$

$$\Rightarrow 1\left(12 + n^2\right) - 1\left(6 + n\right) + n\left(2n - 4\right) = \pm 158$$

$$\Rightarrow 3n^2 - 5n - 152 = 0 \text{ or } 3n^2 - 5n + 164 = 0$$

($\because$ $D < 0$, no real roots of $3n^2 - 5n + 164 = 0$)

$n = 8, \ -\frac{19}{3} \ \Rightarrow \ n = 8$, as $n \geq 0$

then, $\vec{\mathrm{b}} \cdot \vec{\mathrm{c}} = 2 + 4\mathrm{n} - 3\mathrm{n} = 10$

$\vec{\mathrm{a}} \cdot \vec{\mathrm{c}} = 1 + n + 3n = 33$

Q142. Solution

Correct Answer: (A)

$f(x) = x^2 + \int_0^x e^{-t} f(x-t)dt$

$= x^2 + \int_0^x e^{-(x-t)} f(t)dt = x^2 + e^{-x} \int_0^x e^t f(t)dt$

$\therefore \quad e^x f(x) = x^2 e^x + \int_0^x e^t f(t)dt \qquad \ldots(1)$

Differentiating both sides w.r.t.x

$e^x f(x) + e^x f'(x) = 2xe^x + x^2 e^x + e^x f(x)$

$\therefore \quad f'(x) = 2x + x^2$

$\therefore \quad f(x) = x^2 + \frac{x^3}{3} + c$

From (1), we have $f(0) = 0$. So $c = 0$

Hence, $f(x) = x^2 + \frac{x^3}{3} \Rightarrow \sum\limits_{k=1}^{9} f(k) = \Sigma\left(k^2 + \frac{k^3}{3}\right) = 960$